

Executive Summary

Cannabis E-Commerce Hybrid Recommendation System

Most cannabis platforms show the same products to every user. This project builds a personalization engine for a cannabis e-commerce platform that learns from what users do and what products contain, generating recommendations tailored to individual browsing behavior and product attributes like strain type, potency, and effect tags.

The Core Finding from the Data

With 96% of users who clicked on a product adding it to their cart, and 94.6% of those going on to purchase, the platform does not have a conversion problem. It has a discovery problem. Users who find what they want almost always buy it. The recommendation engine addresses the upstream challenge of getting the right product in front of the right user in the first place.

How the Recommendation Engine Works

Two signals are combined into a single hybrid score:

The behavioral signal uses Node2Vec graph embeddings to learn from user-product interaction patterns. Users who engage with similar products get similar recommendations. Interactions are weighted by type so a purchase counts more than a click.

The content signal uses TF-IDF embeddings on product names, descriptions, strain types, and effect tags to measure product similarity. This ensures new or niche products with limited interaction history still surface for relevant users based on what they are, not just how popular they are.

Neither signal alone is enough. Behavior without content misses product relevance. Content without behavior ignores demonstrated user preference. The hybrid approach captures both.

Purchase Prediction

Logistic Regression, Random Forest, and XGBoost classifiers were built to predict purchase likelihood. All three models returned near-random performance. This reflects the simulation design where purchase outcomes were assigned randomly rather than tied to product or user features. In a production dataset where purchases correlate with price sensitivity, brand loyalty, and prior behavior, these models would be expected to perform meaningfully better. Recall remains the right metric to prioritize since missing a potential buyer carries higher cost than an incorrect recommendation.

What This Means for the Business

Concentrates drove the most purchases despite fewer clicks than edibles, suggesting higher purchase intent per session and a potential opportunity to increase concentrate visibility. Vape cartridges had both the fewest interactions and the lowest average ratings, pointing to an assortment or quality gap worth investigating. Flowers had the highest average rating at 3.68, making them strong candidates for featuring in recommendation surfaces targeting new users.

Next Steps

Bias the simulation probabilities by product features to create learnable purchase patterns, integrate user demographics for deeper personalization, and experiment with graph neural networks and LLM-based embeddings as upgrades to the current Node2Vec and TF-IDF architecture.